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Edge-Based Multimodal Driver Monitoring System on Raspberry Pi

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***Abstract*—We present a compact multimodal driver monitoring prototype built on Raspberry Pi for edge-side driver-state awareness. The system integrates a camera on a two-axis gimbal, an inertial sensor, gas sensors for alcohol and smoke cues, and a photoplethysmography sensor for heart rate and SpO_2 estimation. Rather than treating these components as isolated modules, we organize them into a two-stage pipeline: pre-drive readiness screening from physiological and environmental sensing, and in-drive monitoring from visual and motion cues. This poster paper emphasizes the system design, modality coordination, and early proof-of-concept results. Preliminary studies show end-to-end feasibility on-device, including active face tracking, monotonic gas-sensor responses in smoking and BAC-oriented calibration scenarios, interpretable physiological alerts, and a lightweight visual model that reaches 0.89 accuracy in four driving-quality bands on our collected dataset. The prototype illustrates how low-cost edge hardware can support integrated, in-cabin safety monitoring.**

***Index Terms*—Driver monitoring system, Multimodal sensing, Edge computing, Raspberry Pi, In-cabin monitoring, Driving safety.**

I. Introduction

Driver risk inside the vehicle is distributed across several signals. Visual cues reflect fatigue and attention, motion patterns reflect abrupt maneuvers, environmental cues indicate smoking or alcohol exposure, and physiological signals reflect short-term readiness. Many driver monitoring solutions still emphasize a single modality or depend on heavier infrastructure, which makes low-cost in-cabin deployment harder [1], [2].

This work explores a different point in the design space: a self-contained Raspberry Pi prototype that coordinates these complementary cues on-device [3]. The goal is not to deliver a medical or legal judgement, but to provide a practical edge platform that summarizes safety-relevant driver states before and during a drive.

This poster makes three contributions. First, it presents an integrated hardware prototype that combines camera, IMU, gas, and PPG sensing on a single edge platform. Second, it organizes these modalities into a coherent monitoring pipeline rather than a list of independent sensors. Third, it reports preliminary evidence that the prototype supports end-to-end driver-state awareness, including active face tracking, motion-pattern scoring, and readiness alerts from gas and physiological measurements.

II. System Overview

Fig. 1 summarizes the prototype. A Raspberry Pi serves as the acquisition, control, and inference node. The visual module uses a Raspberry Pi Camera V3 NOIR mounted on a PCA9685-driven two-axis gimbal with two SG90 servos, enabling active face-centered capture for fatigue- and attention-related analysis. The motion module uses an MPU6050 to record three-axis acceleration and gyroscope data, which capture abrupt maneuvers such as hard braking, fast starts, and turning. For in-cabin environmental sensing, TGS2620 targets alcohol vapor and MQ135 provides smoke-related gas cues through an MCP3008 ADC. The MAX30102 module measures heart rate and SpO_2, and a KY040 rotary encoder with the local display provides lightweight control over sensing modes.

Each modality covers a different safety dimension. Vision focuses on face visibility and behavior. Motion reflects driving dynamics. Gas sensing provides environmental risk cues. PPG reflects physiological readiness. This division is central to the contribution of the poster: multimodality is used to broaden driver-state awareness rather than to duplicate the same signal with multiple sensors.

The software stack follows a multithreaded design on Raspberry Pi. Sensor acquisition, camera capture, gimbal control, and local interaction run concurrently and feed shared buffers for on-device interpretation. This architecture keeps the prototype responsive while retaining sensing and decision logic locally.

III. Multimodal Pipeline

We organize the system into two decision stages, shown in Fig. 3. Stage 1 performs readiness screening before or at the beginning of a driving session. Gas readings are calibrated into alcohol- and smoke-related indicators, while the RED/IR PPG stream is filtered and summarized to estimate heart rate and SpO_2. These modules are used conservatively: they flag potentially unsafe start conditions instead of making medical or forensic claims.

Stage 2 performs in-drive monitoring. The camera stream is coupled with gimbal control to keep the driver's face within view, allowing the visual module to support active face tracking rather than passive capture alone. Stabilized frames are processed with MediaPipe-based facial landmark analysis [4], [5]. Short-window statistics from the facial

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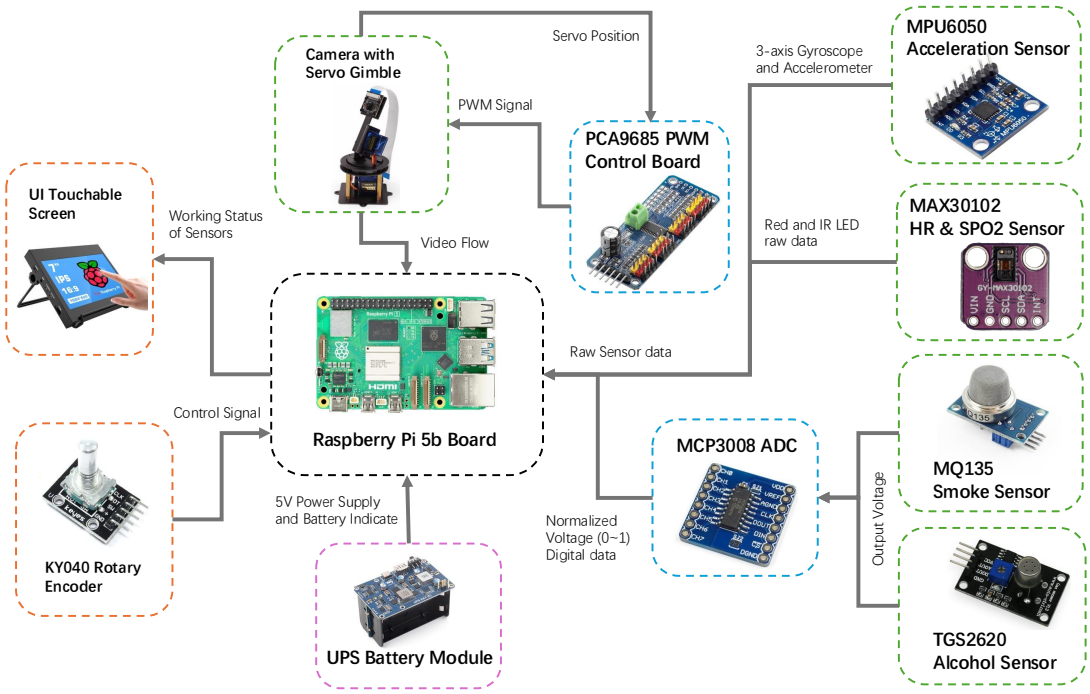
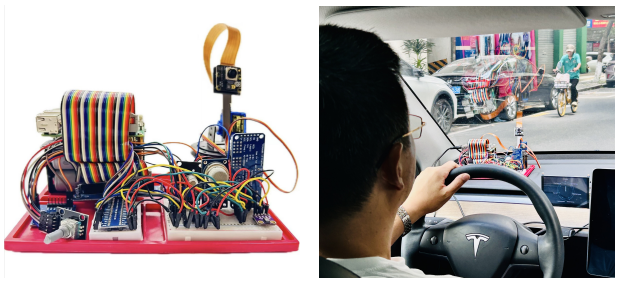


Fig. 1. Poster-scale overview of the Raspberry Pi prototype. The system combines visual, motion, environmental, and physiological sensing into one edge-side driver monitoring loop.



(a) Prototype deployment. (b) Local rotary interaction.

Fig. 2. The portable hardware prototype and its local interface.

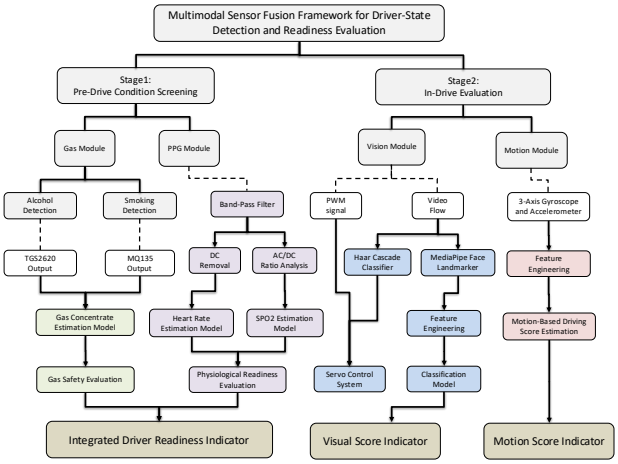


Fig. 3. Two-stage multimodal pipeline: readiness screening before driving, and continuous monitoring during driving.

signals are then mapped to coarse driving-quality bands with a lightweight classifier. In parallel, the IMU stream is transformed into temporal-difference features and summary statistics that capture abrupt steering, braking, and acceleration patterns.

Fusion occurs at the decision level. Gas and PPG modules produce readiness flags; vision and motion modules provide ongoing behavioral evidence; and the local interface exposes status and module selection. The result is an integrated awareness signal rather than a single monolithic classifier. This design keeps each modality interpretable while still treating the system as one coordinated edge prototype.

IV. Preliminary Results and Demonstration

We report early-stage validation rather than a full-scale evaluation. The prototype was exercised in three forms: end-to-end demonstrations on Raspberry Pi, calibration studies for gas and physiological modules, and a collected driving dataset for the visual and motion modules. In live use, the gimbaled camera maintained face-centered capture and the rotary interface supported local control of sensing modes.

For visual and motion analysis, we recorded 40 driving

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TABLE I
POSTER-SCALE EVIDENCE FROM THE PROTOTYPE.

Module	Preliminary finding
Vision	Best four-band driving-quality prediction reached 0.89 accuracy after PCA-based reduction from 156 to 20 features.
Motion	Lower driving-score bands exhibited larger IMU variation, supporting vote-based motion scoring for abrupt maneuvers.
Gas	Smoking and BAC-oriented trials produced separable and monotonic sensor responses.
PPG	Exercise trials raised HR above 120 bpm and occasionally lowered SpO_2 below 94%, motivating readiness alerts.

sessions from 10 drivers on the same road segment. Sessions were grouped into four score bands (65 ± 5, 75 ± 5, 85 ± 5, and 95 ± 5) using Tesla's internal score with cross-checking from AutoBeacon. After reducing the visual representation from 156 dimensions to 20 principal components, the best lightweight classifier (SVM) reached 0.89 accuracy in four-band prediction. For the motion module, lower score bands showed larger acceleration and gyroscope variation than higher score bands, which motivated the rule-based motion voting ranges used in the prototype.

For readiness screening, smoking experiments covered seven in-cabin scenarios and alcohol tests covered 11 BAC-oriented conditions. In both cases, the transformed gas-sensor outputs increased in an interpretable, monotonic manner as risk conditions intensified. Physiological calibration with 10 healthy adults across controlled exercise conditions showed heart-rate increases above 120 bpm after longer exertion and occasional post-exercise SpO_2 values below 94%, supporting conservative alert thresholds for temporary readiness warnings [6], [7].

These results are promising but still preliminary. The main value of the current prototype is end-to-end integration: the system can collect, interpret, and present multiple in-cabin cues on a low-cost edge device without treating any single modality as sufficient on its own.

V. DISCUSSION AND CONCLUSION

This poster reframes the project as an integrated edge prototype for in-cabin driver-state awareness. Its contribution is system-level: it shows how visual tracking, motion sensing, gas cues, and physiological signals can be coordinated on Raspberry Pi to provide broader coverage than any single sensing channel.

The current prototype remains early stage. The evaluation is limited in scale, several thresholds come from controlled calibration settings, and performance may vary with lighting, occlusion, sensor placement, cabin ventilation, and user movement. Continuous visual and physiological monitoring also raises privacy and consent questions that should be addressed with local processing and minimal retention of raw data.

Future work will strengthen robust visual analysis on the edge, test the system in more diverse on-road conditions, and refine privacy-preserving summaries for practical deployment. Even at this stage, the prototype demonstrates that multimodal, edge-side driver monitoring can be organized as a coherent and discussable poster contribution rather than as a collection of disconnected sensing modules.

Ethical Statement: All participants provided informed consent, and no personally identifiable data were retained in the reported analysis.

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